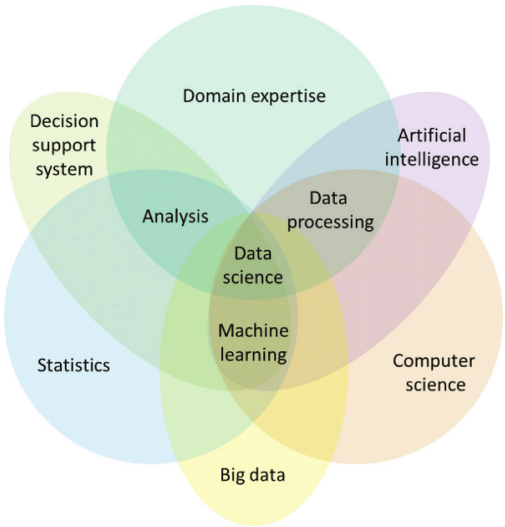
1 Introduction to Statistics

1.1 Statistics

Researchers in the life sciences analyze **data** that exhibit **variability**. For example, patients given the same treatment may respond differently; PCR (Polymerase Chain Reaction) assays prepared identically may yield different results due to contamination, thermal damage, etc. The challenge for the life scientist is to discern the **signal** from the **noise**.

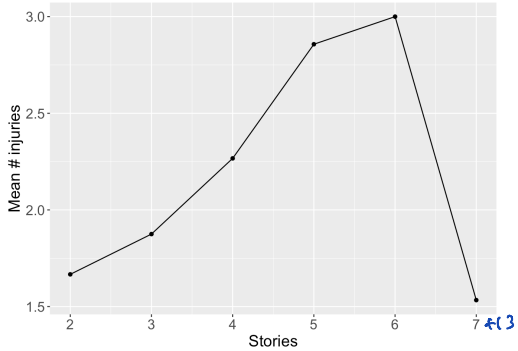
Statistics is the science of understanding data (collection, analysis, and presentation) and of making decisions in the face of variability and uncertainty. Statistics is a core component of **data science**.



Data Science, credit: Lee et al. 2018

Example: Cats in New York

An article presented results on the number of injuries in cats that had plummeted from buildings in New York City, by the number of floors they had fallen. The damage caused by such falls was dubbed *feline high-rise syndrome (FHRS)*.

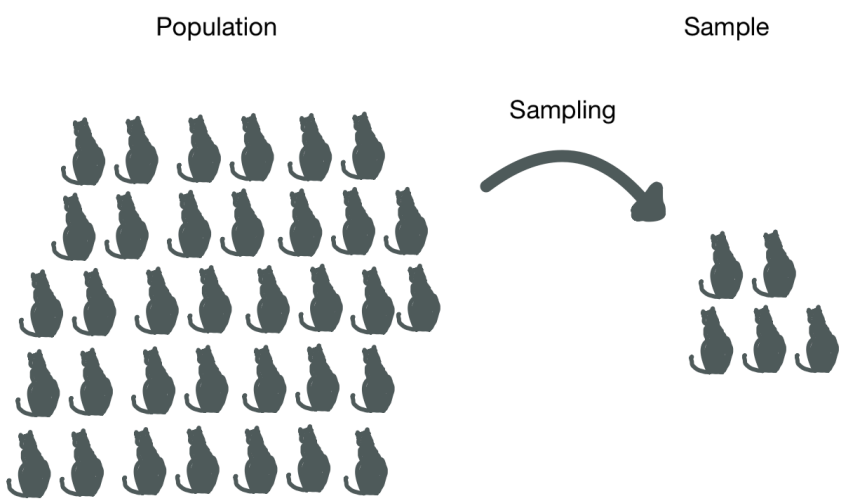


Raining cats

What can we conclude?

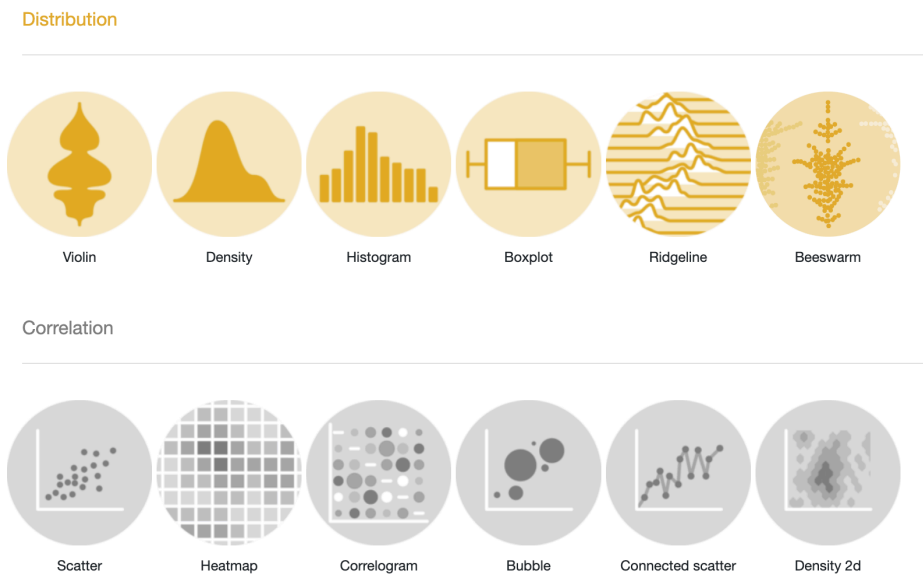
High-quality samples are essential for sound science, even in the age of AI!

Our ability to obtain reliable estimates of population quantities of interest, and to assess uncertainty, depends on how we **sample** populations. It is often at this early step in an investigation that the fate of a study is sealed, for better or worse.



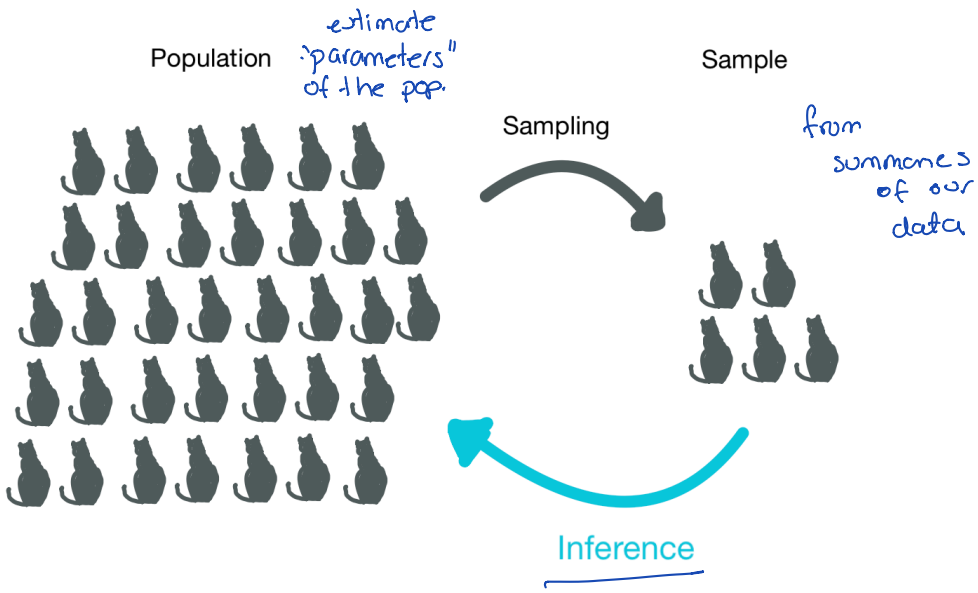
Statistics

In most situations, we do not observe the whole population. Instead, we obtain a sample. We need to ask ourselves: what do we want to do with the data? If our goal is to describe the sample data, we compute **descriptive statistics** and perform an **Exploratory Data Analysis (EDA)**.



EDA https://r-graph-gallery.com/

If instead we want to make statements about the whole population, then beyond EDA we do **statistical inference**.



Statistical Inference

1.2 Data

Your dataset (tabular data) is a collection of variables and observations—typically, one observation per row and one variable per column—filled in with values. A **variable** is a characteristic or quantity that you can measure; we will denote variables with capital letters X, Y, etc. A **value** is the state of the variable when you measure it; we will use lowercase letters (e.g., x, y) for values. An **observation** is a set of measurements from one unit (e.g., one person or one animal).

In the cats example, X = number of injuries is a variable, while x = 2 is one particular value. Similarly, Y = number of floors (stories) fallen is a variable, while y = 5 is a particular value. An observation is the data collected for a particular cat, e.g., Luna fell from the fifth floor, resulting in 2 injuries.

Variables can be **categorical** or **numerical**. Categorical variables indicate membership or attributes. Categorical variables whose categories can be ranked in a meaningful order are called **ordinal**. of cartesorical Examples include blood type (A, B, AB, O), survival status (alive or dead), and language. ex age Numerical variables record the amount of something, for example body temperature, age, or number of friends. Numerical values can be **discrete** (number of friends) or **continuous** (body temperature).

The most common languages used for analyzing data are SAS, R (and Bioconductor), and Python. In this course, we will use R with datasets in the form of **data frames** or **tibbles**.

Let's generate some data by answering this poll.

poll

We can now read our dataset (poll results).

```
#library(readr)

#url <- "https://raw.github.com/JuliaPalacios/stats141_2026/data
#survey <- read_csv(url)
#survey
library(palmerpenguins)
penguins
```

head (penguins)

```
# A tibble: 344 × 8
   species island    bill_length_mm bill_depth_mm
flipper_length_mm body_mass_g
   <fct>   <fct>              <dbl>         <dbl>
<int>       <int>
 1 Adelie  Torgersen           39.1          18.7
181        3750
 2 Adelie  Torgersen           39.5          17.4
186        3800
 3 Adelie  Torgersen           40.3          18
195        3250
 4 Adelie  Torgersen           NA            NA
NA          NA
 5 Adelie  Torgersen           36.7          19.3
193        3450
```

variables

```
 6 Adelie  Torgersen           39.3          20.6
190        3650
 7 Adelie  Torgersen           38.9          17.8
181        3625
 8 Adelie  Torgersen           39.2          19.6
195        4675
 9 Adelie  Torgersen           34.1          18.1
193        3475
10 Adelie  Torgersen           42            20.2
190        4250
# ℹ 334 more rows
# ℹ 2 more variables: sex <fct>, year <int>
```

A first step is to explore the data and describe them in summary form. Common checks include:

- Measures of center and measures of dispersion
- Checking for missing values (e.g., "NAs")
- Plotting data, frequency distributions, and charts
- Talking with the person who collected the data (review the data dictionary)

```
summary(penguins$species)
```

```
   Adelie Chinstrap    Gentoo
      152        68       124
```

```
summary(penguins$flipper_length_mm)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  172.0   190.0   197.0   200.9   213.0   231.0       2
```

```
#mean(penguins$flipper_length_mm, na.rm = TRUE)  #ignore NA
```

↑ removes NAs for mean computation.

Measures of center (or typical value):

- Sample mean (average): ($\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$)
- Median: the value that lies in the middle of the ordered sample
- Mode: the value (or values) that occurs most frequently in a dataset

Measures of dispersion: / spread.

- Range: max − min
- Interquartile range (IQR): the spread of (roughly) the middle 50% of the observations (25%, 75%)
- Deviation: $y_i - \bar{y}$

mean deviation $= \frac{1}{n}\sum_{i=1}^{n} |y_i - \bar{y}|$

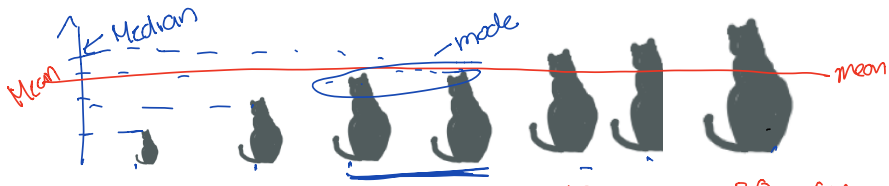
- Sample standard deviation: $s = \sqrt{\frac{\sum_{i=1}^{n}(y_i-\bar{y})^2}{n-1}}$

euclidean distance

- Sample variance: s^2
- Coefficient of variation: cv= $\frac{s \times 100\%}{\bar{y}}$

sq root of average of squared deviations

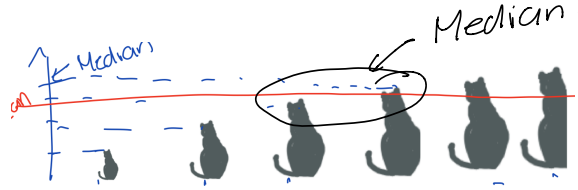
↑ scale free.



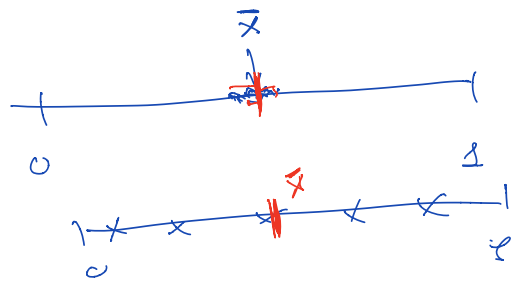
$$\frac{\sum x_i}{n}$$

Median = this cat's height

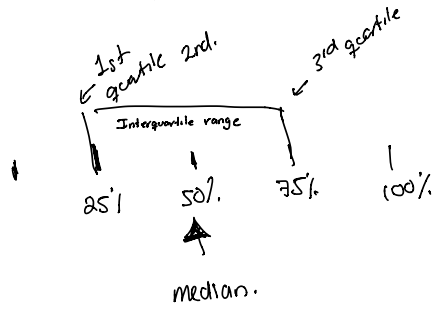
If we have an even number of observations, the median is the average of the values in the middle



Measures of spread / deviation



Quantiles: We divide our <u>ranked</u> data into 4 parts



Why (n-1) in the standard deviation?

$y_i - \bar{y}$

Note that the sum of deviations (y_i-{y}) is always zero. This means that, in a sample of (n) observations, there are only (n-1) independent pieces of information about the deviations from the average. The quantity (n-1) is called the **degrees of freedom** of the standard deviation (and the variance).

$$\sum_{i=1}^{n}(y_i - \bar{y})$$

$$\sum_{i=1}^{n} y_i - \sum_{i=1}^{n} \bar{y}$$

$$\cancel{n}\frac{\sum y_i}{\cancel{n}} - n\bar{y}$$

$$n\bar{y} - n\bar{y} = 0$$

1.3 Plotting the data

We will start with a **scatter plot**.

A scatter plot is a tool for examining the relationship between two numerical variables (X) and (Y). It plots each observed ((x,y)) pair as a dot on the (x)-(y) plane.

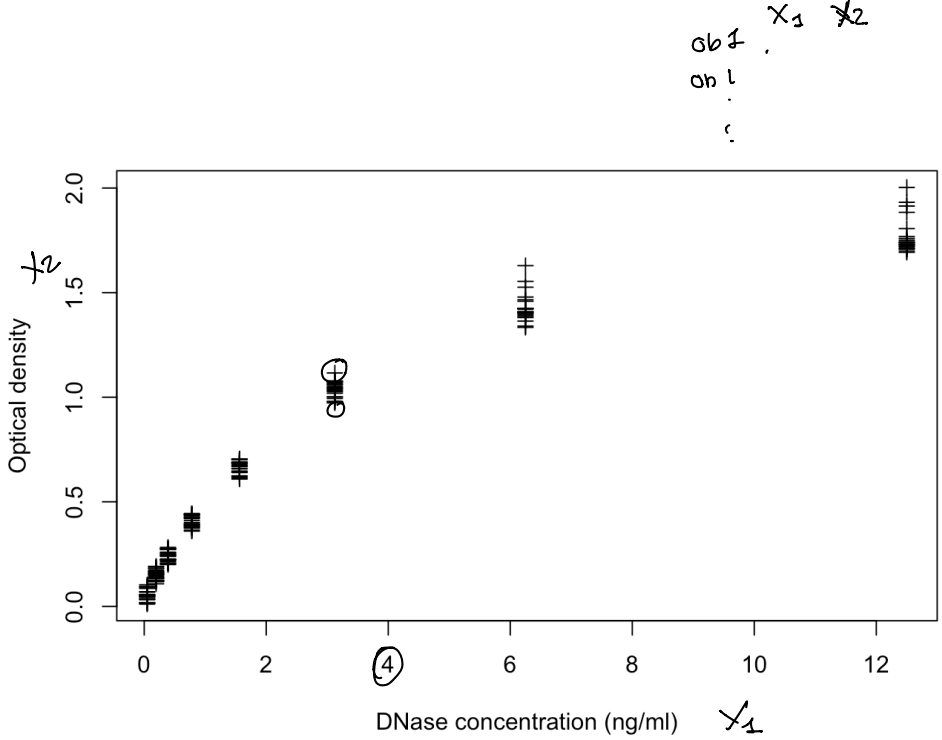
```
?DNase  #what is the dataset about?
head(DNase) #shows the first rows
```

```
  Run       conc density
1   1 0.04882812   0.017
2   1 0.04882812   0.018
3   1 0.19531250   0.121
4   1 0.19531250   0.124
5   1 0.39062500   0.206
6   1 0.39062500   0.215
```

```
names(DNase) # gives you the name of the variables
```

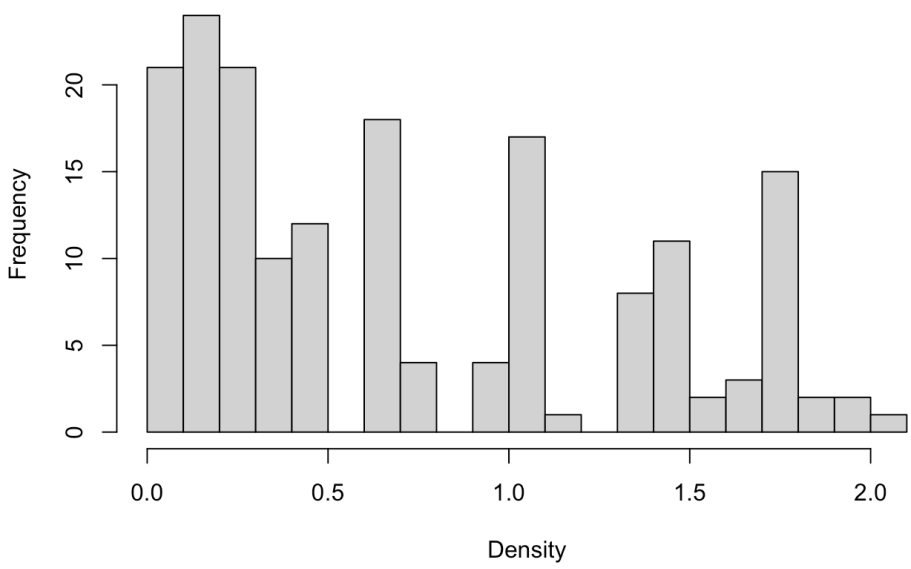
```
[1] "Run"     "conc"    "density"
```

```
plot(DNase$conc, DNase$density,
  ylab = attr(DNase, "labels")$y,
  xlab = paste(attr(DNase, "labels")$x, attr(DNase, "units")$x),
  pch = 3)
```

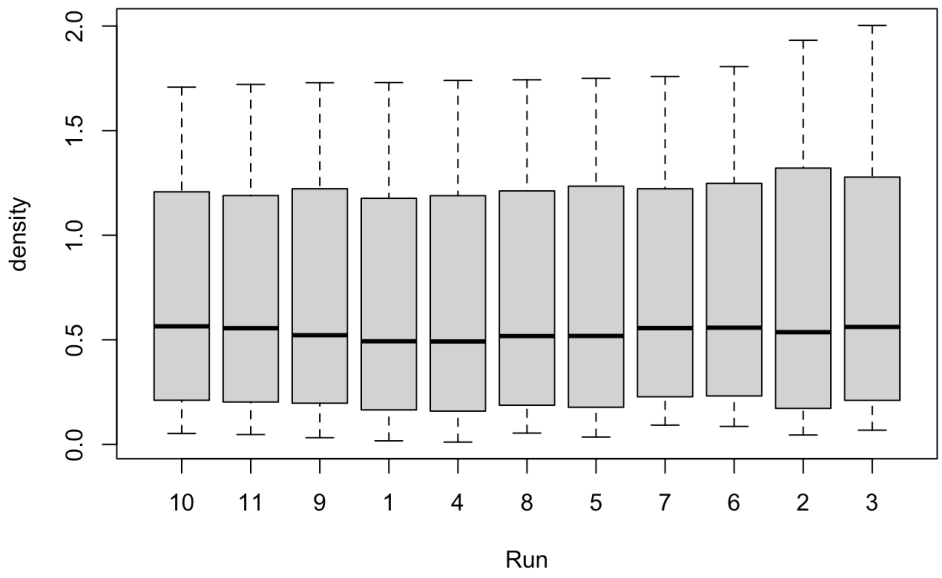


We can visualize *frequency distributions* with **bar plots** (as well as jittered dotplots), **histograms**, and **boxplots**. Bar plots and histograms display the frequency of values (or grouped values). A boxplot displays the minimum, 1st quartile, median, 3rd quartile, and maximum at once. We can also compare multiple distributions at once.

```
hist(DNase$density, breaks=25, main = "",xlab="Density")
```




```
boxplot(density ~ Run, data = DNase)
```



The **shape** of a distribution is often informative. It may be, for example: (a) symmetric and bell-shaped, (b) symmetric but not bell-shaped, (c) right-skewed, (d) left-skewed, (e) exponential, or (f) bimodal.

Base R plotting functions are good for quick exploration, but they have aesthetic limitations. We will use a framework called the **grammar of graphics**, implemented in the package **ggplot2**.